Regular Languages

A brief introduction to alphabets, strings and languages

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1 Some basic definitions

Here I introduce some very basic words.

Definition 1.1 (Alphabet, string).

The following are two fundamental building blocks of formal language theory:

1. Alphabet: An alphabet Σ is a finite, non-empty set of symbols, whose elements are called letters.
2. String: A string over an alphabet Σ is a finite ordered sequence of elements of Σ.
   (a) The length of a string w, denoted by $|w|$, is the number of symbols in the sequence.
   (b) The unique string of length zero is called the empty string and is denoted by ε.

Here are some basic examples:

1. Binary alphabet. Consider the alphabet
$$\Sigma = \{0, 1\}.$$
Some strings over this alphabet are
$$0, \quad 1, \quad 001, \quad 1010, \quad 11001.$$
2. English alphabet. Consider the alphabet
$$\Sigma = \{a, b, c, \ldots, z\}.$$
Some strings over this alphabet are
$$\texttt{automata}, \quad \texttt{theory}, \quad \texttt{computer}.$$
3. Decimal alphabet. Consider the alphabet
$$\Sigma = \{0, 1, 2, \ldots, 9\}.$$
Some strings over this alphabet are
$$123, \quad 2026, \quad 314159.$$

Definition 1.2 (Concatenation).

The concatenation of u and v, denoted by uv, is the string obtained by appending the symbols of v to the end of u. In particular, concatenation satisfies

$$|uv| = |u| + |v|,$$

and the empty string ε acts as the identity:

$$u\varepsilon = \varepsilon u = u.$$

The notation Σ^* denotes the set of all finite strings over an alphabet Σ, including the empty string ε.

Let $u, v, w \in \Sigma^*$. The concatenation operation satisfies the following properties:

1. Closure. The concatenation of two strings is again a string over Σ:
$$uv \in \Sigma^*.$$
2. Associativity. Concatenation is associative:
$$(uv)w = u(vw).$$
3. Identity element. The empty string ε acts as the identity element:
$$u\varepsilon = \varepsilon u = u.$$
4. Non-commutativity. Concatenation is not commutative in general:
$$uv \neq vu.$$

It is also important to observe that non-empty strings do not possess inverses under concatenation. This follows immediately from the fact that

$$|uv| = |u| + |v|,$$

which implies that $uv \neq \varepsilon$ whenever $u \neq \varepsilon$. Consequently, the algebraic structure $(\Sigma^*, \cdot)$, where $\cdot$ denotes concatenation, forms a monoid, but not a group.

1.1 Languages and basic operations

Definition 1.3 (Language).

Let Σ be an alphabet. A language L over Σ is any subset of Σ^*, where Σ^* denotes the set of all finite strings over Σ, including the empty string ε. Thus,

$$L \subseteq \Sigma^*.$$

Some examples of languages are given below:

1. Binary language. Consider the alphabet
$$\Sigma = \{0, 1\}.$$
The collection of all binary strings ending in 1 forms a language:
$$L = \{w \in \Sigma^* : w \text{ ends in } 1\}.$$
2. English language. The English language can be regarded as a language over an alphabet consisting of the English letters, spaces, and punctuation marks such as full stops, commas, semicolons, and colons. The collection of valid English expressions forms a language over this alphabet.
3. Empty language. The empty set $\varnothing$ is also a language over any alphabet, since
$$\varnothing \subseteq \Sigma^*.$$

Definition 1.4 (Concatenation of Languages).

Let Σ be an alphabet and let $L_1, L_2 \subseteq \Sigma^*$ be two languages. The concatenation of L_1 and L_2, denoted by L_1L_2, is defined as

$$L_1L_2 = \{uv : u \in L_1,\ v \in L_2\}.$$

In other words, it is the set of all strings obtained by concatenating a string from L_1 with a string from L_2.

Some examples of language concatenation are given below:

1. Let
$$L_1 = \{0, 01\}, \qquad L_2 = \{1, 10\}.$$
Their concatenation is
$$L_1L_2 = \{01, 010, 011, 0110\}.$$
2. Let
$$L_1 = \{a, ab\}, \qquad L_2 = \{b, ba\}.$$
Then
$$L_1L_2 = \{ab, aba, abb, abba\}.$$

The concatenation of languages satisfies the following basic properties. Let $L_1, L_2, L_3 \subseteq \Sigma^*$ be languages.

1. Distributivity over union. Concatenation distributes over union:
$$L_1(L_2 \cup L_3) = L_1L_2 \cup L_1L_3,$$
$$(L_1 \cup L_2)L_3 = L_1L_3 \cup L_2L_3.$$
2. Associativity. Concatenation is associative:
$$(L_1L_2)L_3 = L_1(L_2L_3).$$
3. Identity element. The singleton language $\{\varepsilon\}$ acts as the identity element:
$$L_1\{\varepsilon\} = \{\varepsilon\}L_1 = L_1.$$

Definition 1.5 (Kleene Star).

Let Σ be an alphabet and let $L \subseteq \Sigma^*$ be a language. The Kleene star of L, denoted by L^*, is the set of all strings obtained by concatenating zero or more strings from L. Formally,

$$L^* = \bigcup_{n=0}^{\infty} L^n,$$

where

$$L^0 = \{\varepsilon\}, \qquad L^{n+1} = L^nL.$$

Some examples of the Kleene star operator are given below:

1. Consider the language
$$L = \{a\}.$$
Its Kleene star is
$$L^* = \{\varepsilon, a, aa, aaa, \ldots\}.$$
2. Consider the language
$$L = \{0, 1\}.$$
Then
$$L^* = \{\varepsilon, 0, 1, 00, 01, 10, 11, \ldots\},$$
which is precisely the set of all finite binary strings.

The Kleene star operator satisfies the following basic properties:

1. Empty string. The empty string always belongs to the Kleene star of a language:
$$\varepsilon \in L^*.$$
2. Containment. Every string in L also belongs to L^*:
$$L \subseteq L^*.$$
3. Closure under concatenation. The Kleene star of a language is closed under concatenation. That is, if $x, y \in L^*$, then
$$xy \in L^*.$$
4. Interaction with union. For any two languages L_1, L_2,
$$(L_1 \cup L_2)^* = (L_1^* L_2^*)^*.$$

Remark 1.6

The Kleene star of the empty language is
$$\varnothing^* = \{\varepsilon\}.$$
Indeed, by definition, $L^0 = \{\varepsilon\}$, whereas $L^n = \varnothing$ for every $n \geq 1$ when $L = \varnothing$. Thus, the only string in $\varnothing^*$ is the empty string.